Cost–Volume–Profit Analysis

Basic Assumptions

- Changes in production/sales volume are the sole cause for cost and revenue changes
- Total costs consist of fixed costs and variable costs
- Revenue and costs behave and can be graphed as a linear function (a straight line)
- Selling price, variable cost per unit, and fixed costs are all known and constant
- In many cases only a single product will be analyzed. If multiple products are studied, their relative sales proportions are known and constant
- The time value of money (interest) is ignored

Basic Formulae

		Total		Cost		Pretax
Operating		Revenues	_	of	_	Operating
Income	=	from		Goods		Expenses
		Operations		Sold		

Net		Operating	_	Income
Income	=	Income		Taxes

Contribution Margin

	5 Packages Sold	40 Packages Sold
Revenues	$ 1,000 ($200 per package × 5 packages)	$8,000 ($200 per package × 40 packages)
Variable purchase costs	600 ($120 per package × 5 packages)	4,800 ($120 per package × 40 packages)
Fixed costs	2,000	2,000
Operating income	$(1,600)	$1,200

- The only numbers that change as a result of selling different quantities of packages are *total revenues* and *total variable costs*. The difference between total revenues and total variable costs is called **contribution margin**. That is,

$$\text{Contribution margin} = \text{Total revenues} - \text{Total variable costs}$$

- **Contribution margin per unit** is a useful tool for calculating contribution margin and operating income. It is defined as:

$$\text{Contribution margin per unit} = \text{Selling price} - \text{Variable cost per unit}$$

Contribution Margin

- Contribution Margin also equals contribution margin per unit multiplied by the number of units sold

$$\text{Contribution margin} = \text{Contribution margin per unit} \times \text{Number of units sold}$$

- When companies sell multiple products, calculating contribution margin per unit is cumbersome. Instead of expressing contribution margin in dollars per unit, these companies express it as a percentage called **contribution margin percentage** (or **contribution margin ratio**):

$$\text{Contribution margin percentage (or contribution margin ratio)} = \frac{\text{Contribution margin}}{\text{Revenues}}$$

Expressing CVP Relationships

There are three related ways (we will call them “methods”) to model CVP relationships:

1. The equation method
2. The contribution margin method
3. The graph method

Equation Method and Contribution Margin

	5 Packages Sold	40 Packages Sold
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Fixed costs	2,000	2,000
Operating income	$(1,600)	$1,200

$$\left[\begin{pmatrix}\text{Selling}\\\text{price}\end{pmatrix}\times\begin{pmatrix}\text{Quantity of}\\\text{units sold}\end{pmatrix}-\begin{pmatrix}\text{Variable cost}\\\text{per unit}\end{pmatrix}\times\begin{pmatrix}\text{Quantity of}\\\text{units sold}\end{pmatrix}\right]-\begin{matrix}\text{Fixed}\\\text{costs}\end{matrix}=\begin{matrix}\text{Operating}\\\text{income}\end{matrix}\qquad\text{(Equation 1)}$$

$$\left[\begin{pmatrix}\text{Selling}\\\text{price}\end{pmatrix.-\left.\begin{matrix}\text{Variable cost}\\\text{per unit}\end{matrix}\right)\times\begin{pmatrix}\text{Quantity of}\\\text{units sold}\end{pmatrix}\right]-\begin{matrix}\text{Fixed}\\\text{costs}\end{matrix}=\begin{matrix}\text{Operating}\\\text{income}\end{matrix}$$

$$\begin{pmatrix}\text{Contribution margin}\\\text{per unit}\end{pmatrix.\times\left.\begin{matrix}\text{Quantity of}\\\text{units sold}\end{matrix}\right)-\begin{matrix}\text{Fixed}\\\text{costs}\end{matrix}=\begin{matrix}\text{Operating}\\\text{income}\end{matrix}$$

Graph Method

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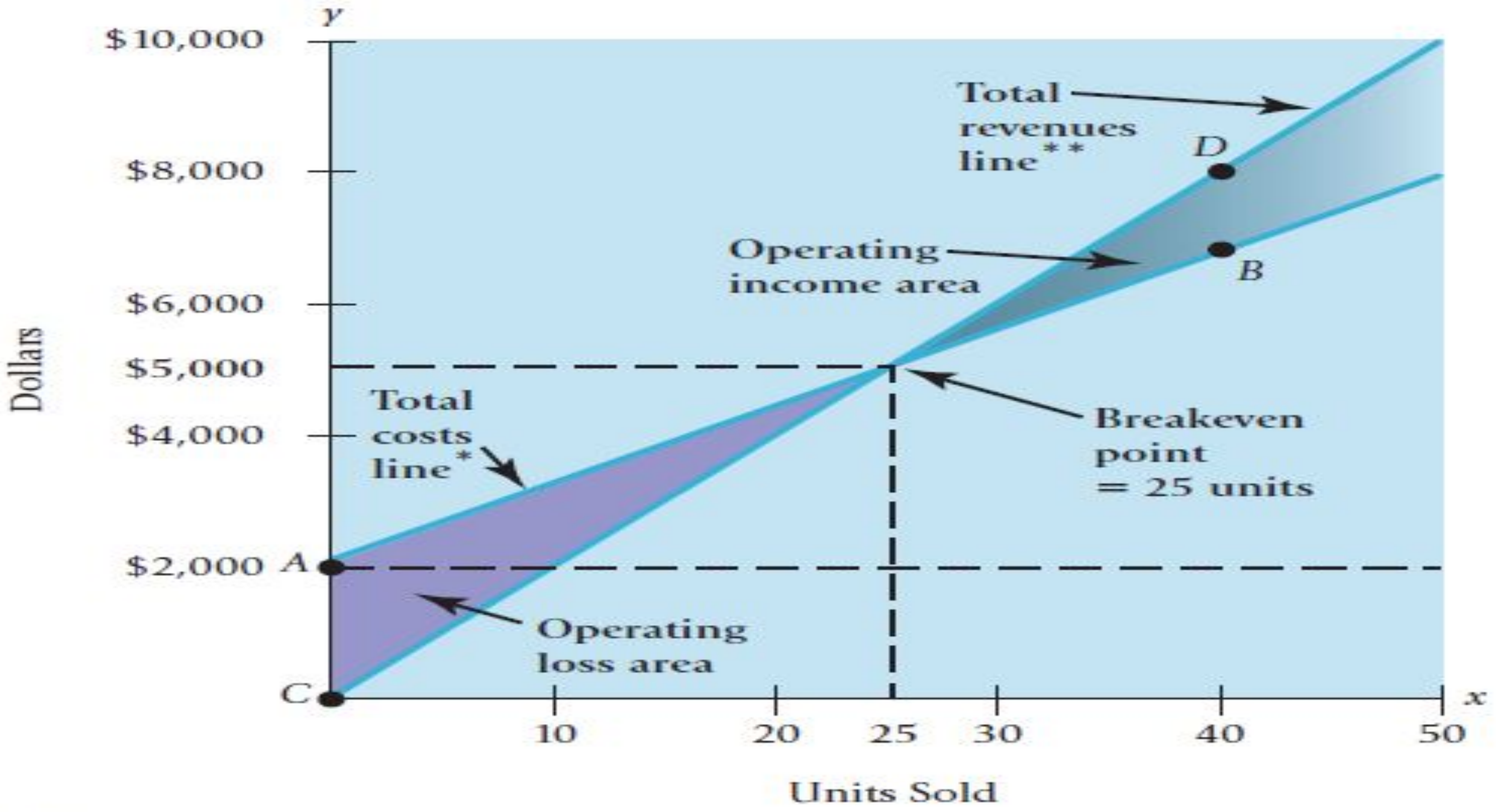


*Slope of the total costs line is the variable cost per unit = $120
**Slope of the total revenues line is the selling price = $200

Break-even analysis

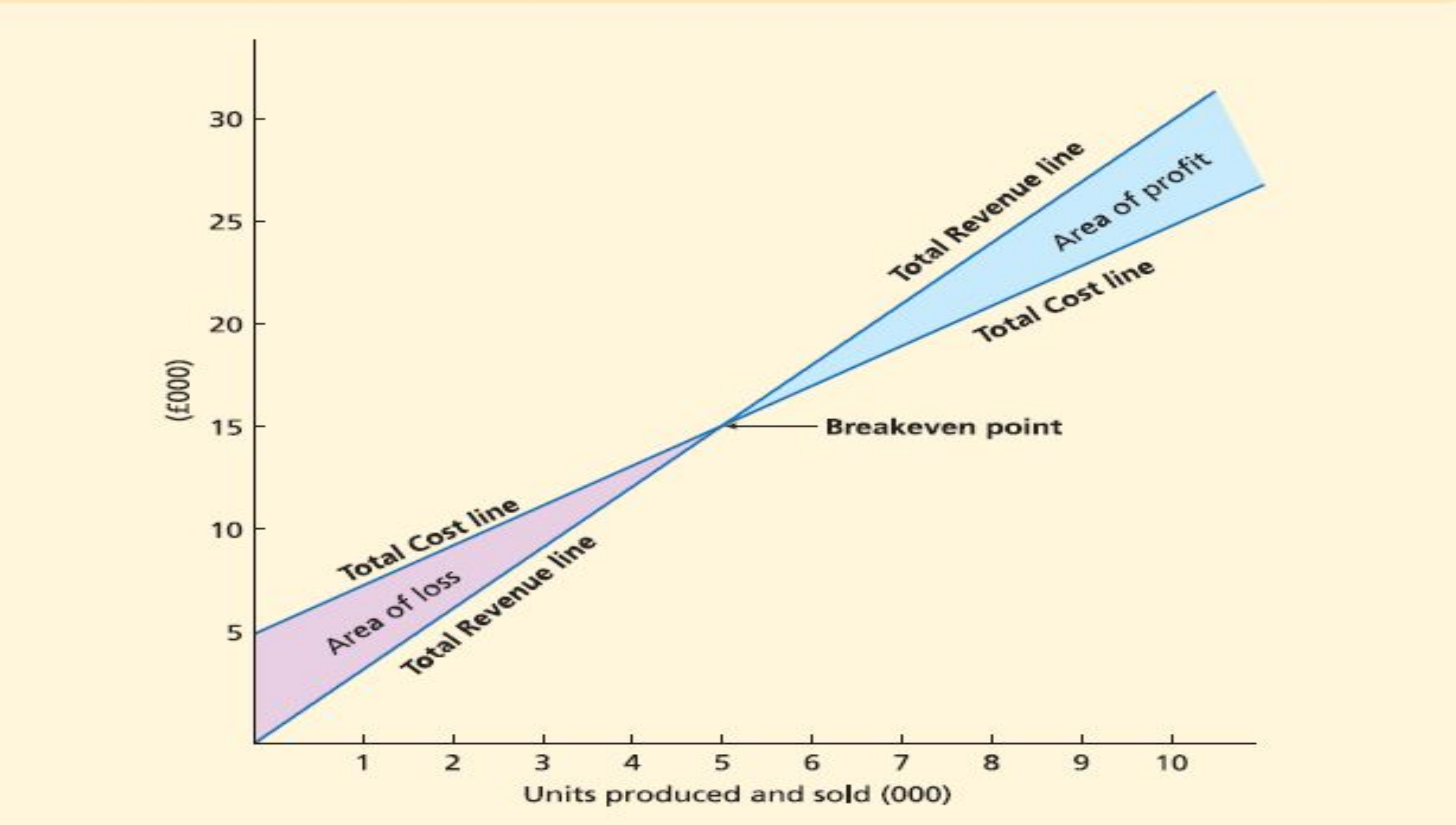
- **Fixed costs stay unchanged over stated ranges in the volume of production, but variable costs change *in total* when the volume of production changes within a stated range.**
- As revenue increases so do variable costs, so that the only item that remains unchanged is that of fixed costs.

$$\text{Breakeven point} = \frac{\text{Fixed costs}}{\text{Selling price per unit} - \text{Variable costs per unit}}$$

Break-even analysis

No. of units	Fixed cost	Variable cost	Total cost: Variable + Fixed	Revenue (Sales)	Profit	Loss
	£	£	£	£	£	£
100	2,000	400	2,400	900		1,500
200	2,000	800	2,800	1,800		1,000
300	2,000	1,200	3,200	2,700		500
400	2,000	1,600	3,600	3,600	nil	nil
500	2,000	2,000	4,000	4,500	500	
600	2,000	2,400	4,400	5,400	1,000	
700	2,000	2,800	4,800	6,300	1,500	
800	2,000	3,200	5,200	7,200	2,000	
900	2,000	3,600	5,600	8,100	2,500	

The breakeven chart



The breakeven chart

- As production and sales go above 5,000 units, the firm makes profits. When production and sales are above 5,000 units, the difference represents the **margin of safety.**
- This is the number of units in excess of the breakeven point. If volume fell by more than the margin of safety, the business would incur losses.

Changes and breakeven charts

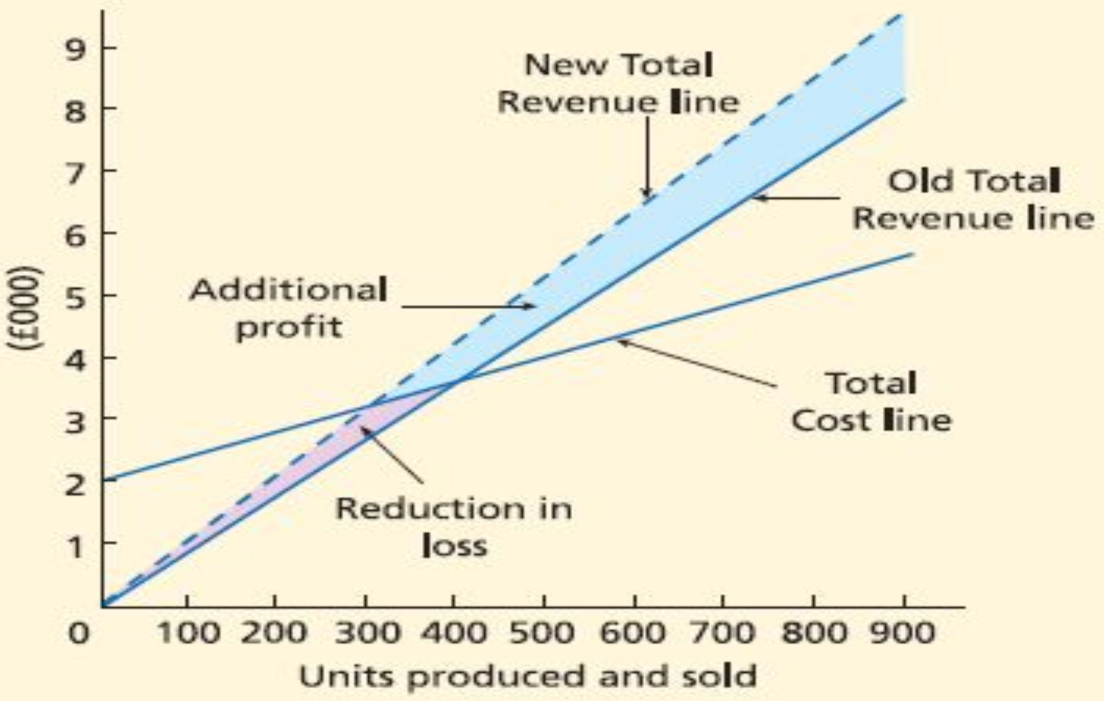
- The effect of changes on profits can easily be shown by drawing fresh lines on the chart to show the changes, or intended changes, in the circumstances of the firm. Let’s first consider some factors that can bring about a change in profits:

- the selling price per unit could be increased (or decreased);
- a possible decrease (or increase) in fixed costs;
- a possible decrease (or increase) in variable costs per unit;
- increase the volume of production and sales.

(a) Increase selling price

- Let us suppose that the selling price could be increased by £2 per unit.

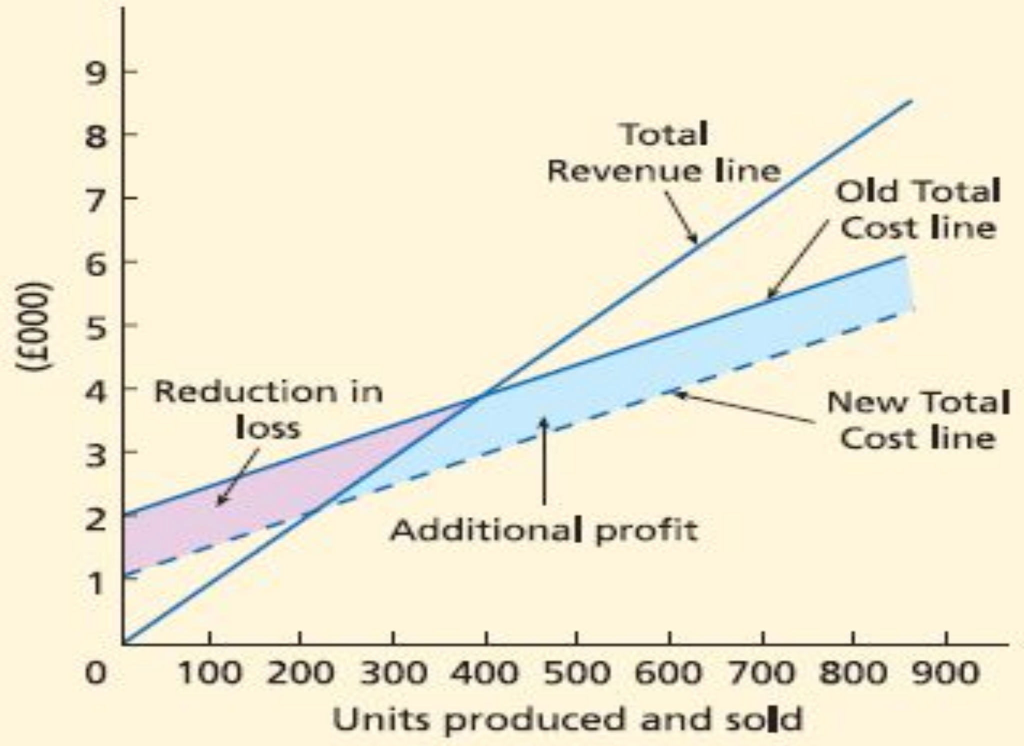
Exhibit 49.6



(b) Reduce fixed costs

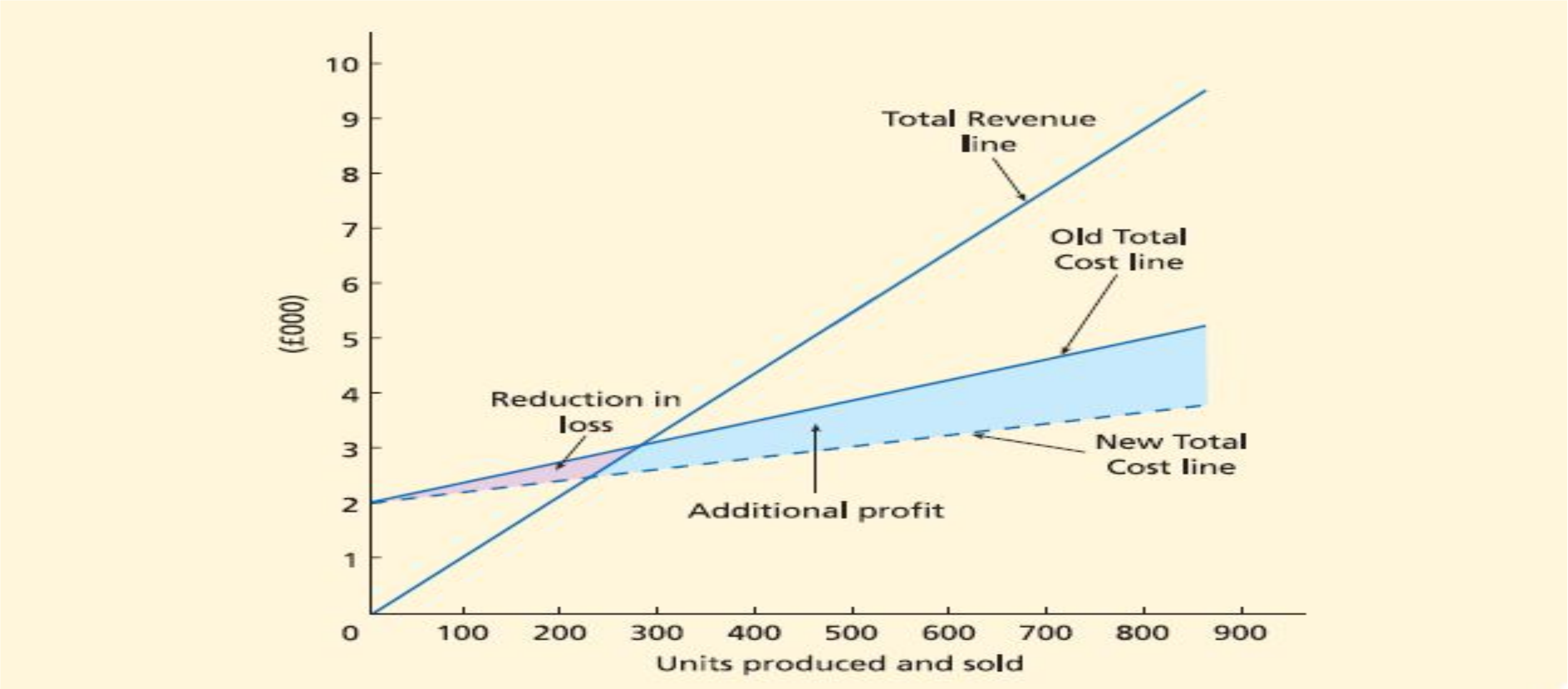
- This time to reflect a reduction of £800 in fixed costs.

Exhibit 49.7



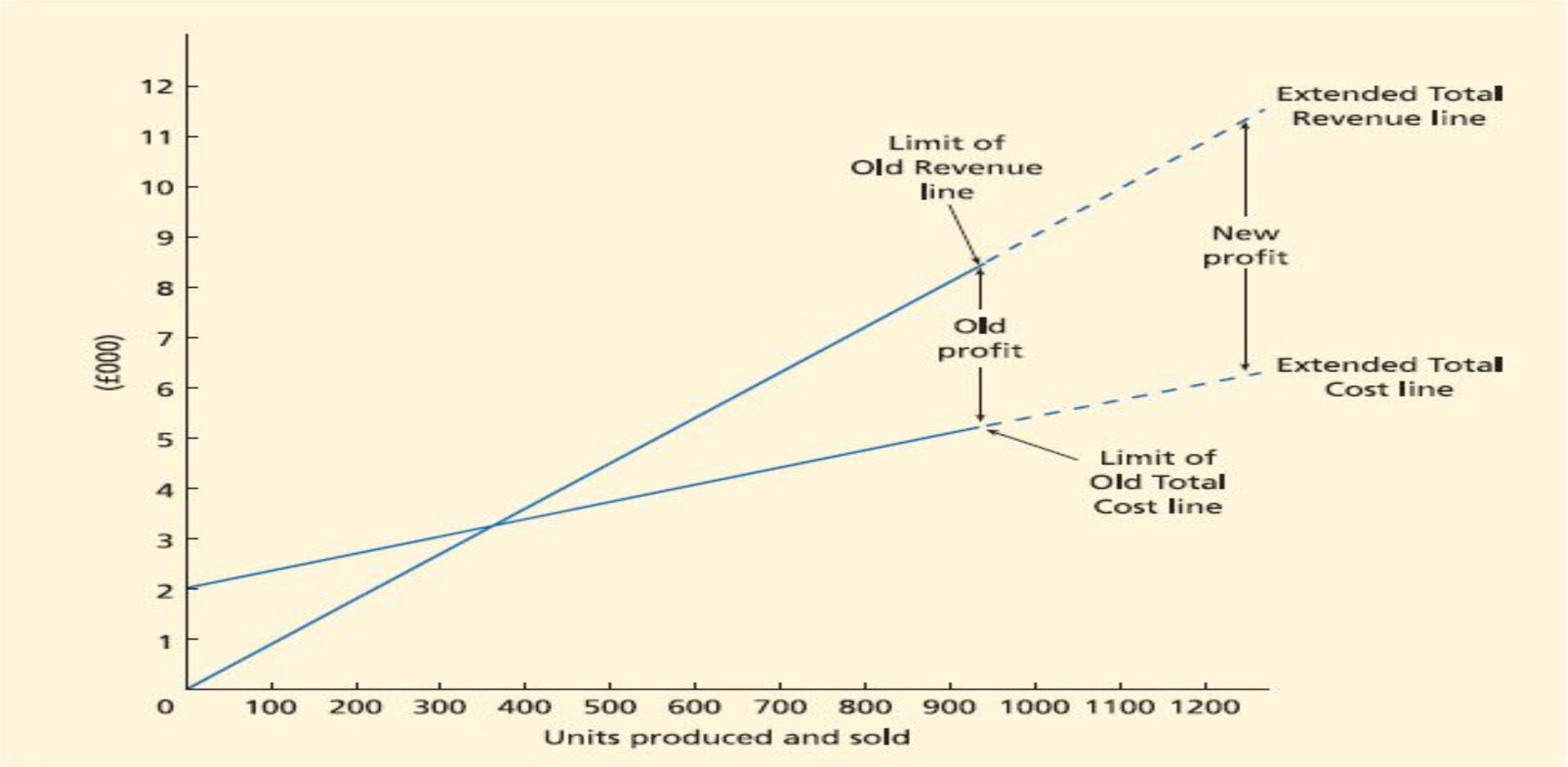
(c) Reduce variable costs

- we can see what happens if the variable costs per unit are reduced, in this case by £2 per unit.



(d) Increased production and sales

- when sales and/or production increase, all that is required is that the lines Total Revenue and Total Cost are extended.



The limitations of breakeven charts

- In each of the cases looked at it has been assumed that only one of the factors of variable cost, fixed cost, selling price or volume of sales has altered.
- This is not usually the case. An increase in price may well reduce the number sold. There may well be an increase in fixed cost which has an effect which brings down variable costs. The changes in the various factors should, therefore, be studied simultaneously rather than separately.
- In addition, **where there is more than one product, the proportions in which the products are sold, i.e. the product mix, can have a very important bearing on costs.** Suppose that there are two products, one has a large amount of fixed costs but hardly any variable costs, and the other has a large amount of variable costs but little fixed costs. If the proportions in which each is sold change very much then this could mean that the costs and profit could vary tremendously, even though the total figures of sales stayed constant.

The limitations of breakeven analysis

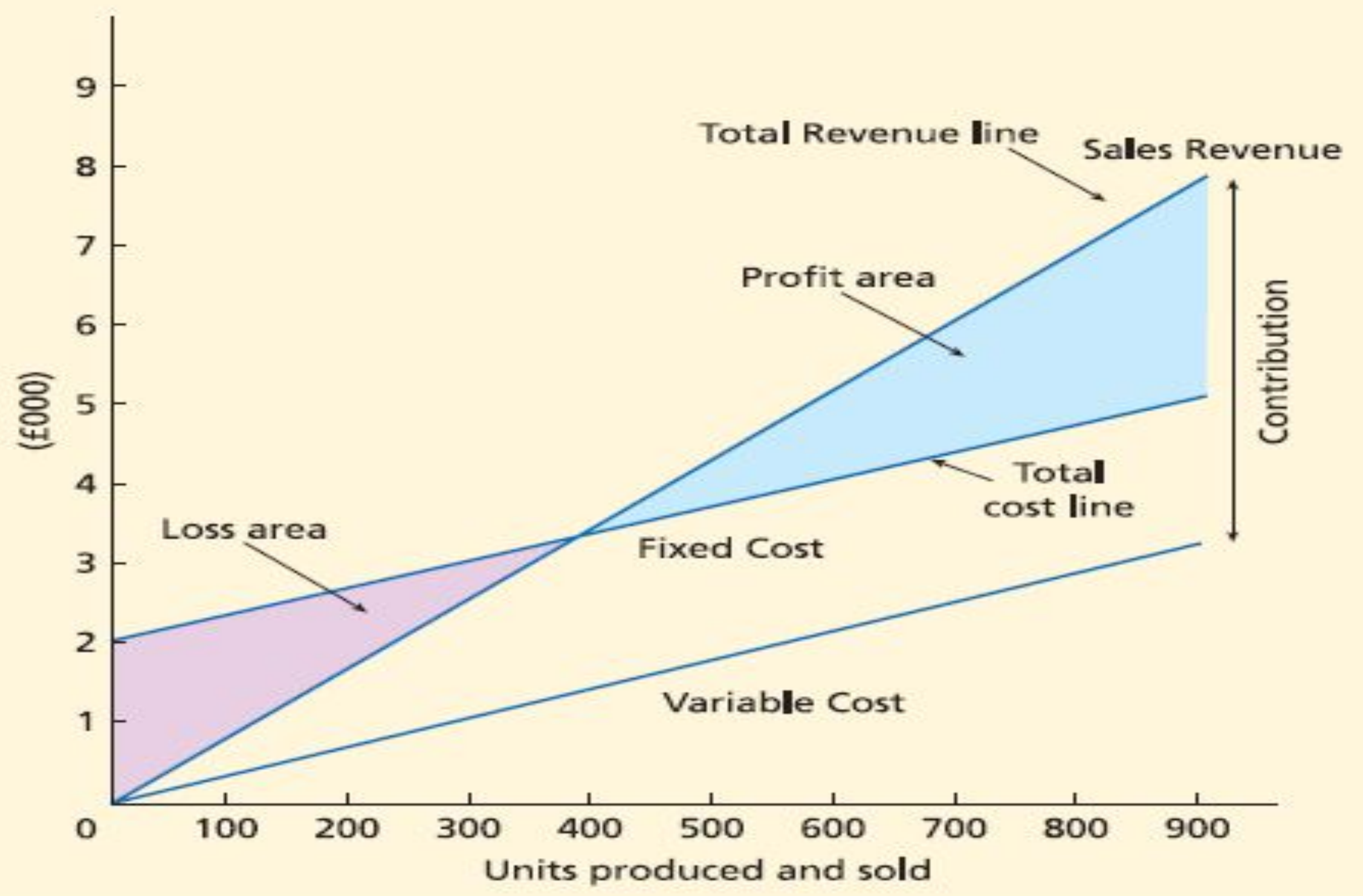
In considering the breakeven analysis we may expect that the following will occur.

Fixed costs £1,000, Variable costs: Product A £5 per unit, B £20 per unit.
Selling prices: A £10 per unit, B £30 per unit.
Expected sales: A 150, B 50. Actual sales: A 30, B 90.

The expected sales are: A 150 × £10 + B 50 × £30 = £3,000.
The actual sales are: A 30 × £10 + B 90 × £30 = £3,000.

Contribution graph

Exhibit 49.11



Using CVP Analysis for Decision-Making.

- A manager can also use CVP analysis to make other strategic decisions such as choosing the product features of engine size, transmission system, or steering system for a new car model.
- Different choices will affect the vehicle’s selling price, variable cost per unit, fixed costs, and units sold.
- CVP analysis helps managers estimate the expected profitability of different choices.

Decision to Advertise

- Suppose Emma anticipates selling 40 units of the *GMAT Success* package at the fair. Exhibit 3-3 indicates that Emma's operating income will be $1,200. Emma is considering advertising the product and its features in the fair brochure. The advertisement will be a fixed cost of $500. Emma thinks that advertising will increase sales by 10% to 44 packages. Should Emma advertise? The following table presents the CVP analysis.

	40 Packages Sold with No Advertising (1)	44 Packages Sold with Advertising (2)	Difference (3) = (2) − (1)
Revenues ($200 × 40; $200 × 44)	$8,000	$8,800	$ 800
Variable costs ($120 × 40; $120 × 44)	4,800	5,280	480
Contribution margin ($80 × 40; $80 × 44)	3,200	3,520	320
Fixed costs	2,000	2,500	500
Operating income	$1,200	$1,020	$ (180)

Decision to Reduce the Selling Price

- Having decided not to advertise, Emma is contemplating whether to reduce the selling price to \$175. At this price, she thinks she will sell 50 units. At this quantity, the test-prep package company that supplies *GMAT Success* will sell the packages to Emma for \$115 per unit instead of \$120. Should Emma reduce the selling price?

Contribution margin from lowering price to \$175: (\$175 − \$115) per unit × 50 units	\$3,000
Contribution margin from maintaining price at \$200: (\$200 − \$120) per unit × 40 units	3,200
Change in contribution margin from lowering price	\$ (200)

Determining Target Prices

- Emma could also ask, "At what price can I sell 50 units (purchased at $115 per unit) and still earn an operating income of $1,200?" The answer is $179, as the following calculations show:

	Target operating income	$1,200
	Add fixed costs	2,000
	Target contribution margin	$3,200
	Divided by number of units sold	÷ 50 units
	Target contribution margin per unit	$ 64
	Add variable cost per unit	115
	Target selling price	$ 179
Proof:	Revenues, $179 per unit × 50 units	$8,950
	Variable costs, $115 per unit × 50 units	5,750
	Contribution margin	3,200
	Fixed costs	2,000
	Operating income	$1,200

Sensitivity Analysis

- **Sensitivity analysis** is a “what if” technique managers use to examine how an outcome will change if the original predicted data are not achieved or if an underlying assumption changes.
- The analysis answers questions such as “What will operating income be if the quantity of units sold decreases by 5% from the original prediction?” and “What will operating income be if variable cost per unit increases by 10%?”
- “What” happens to profit “if”:
  - Selling price changes
  - Volume changes
  - Cost structure changes
    - Variable cost per unit changes
    - Fixed cost changes

Margin of Safety

- An important aspect of sensitivity analysis is **margin of safety**
- A indicator of risk, the Margin of Safety (MOS) measures the distance between budgeted sales and breakeven sales.

$$\text{Margin of safety} = \text{Budgeted (or actual) revenues} - \text{Breakeven revenues}$$

$$\text{Margin of safety (in units)} = \text{Budgeted (or actual) sales quantity} - \text{Breakeven quantity}$$

- The MOS Ratio removes the firm's size from the output, and expresses itself in the form of a percentage:

$$\text{Margin of safety percentage} = \frac{\text{Margin of safety in dollars}}{\text{Budgeted (or actual) revenues}}$$

Operating Leverage

- *Operating leverage* measures the risk-return tradeoff across alternative cost structures.
- **Operating leverage** describes the effects that fixed costs have on changes in operating income as changes occur in units sold and contribution margin.

$$\frac{\text{Degree of}}{\text{operating leverage}} = \frac{\text{Contribution margin}}{\text{Operating income}}$$

Operating Leverage

operating leverage Operating income

The following table shows the degree of operating leverage at sales of 40 units for the three rental options.

	Option 1	Option 2	Option 3
1. Contribution margin per unit (see page 98)	$ 80	$ 50	$ 30
2. Contribution margin (row 1 × 40 units)	$3,200	$2,000	$1,200
3. Operating income (from Exhibit 3-5)	$1,200	$1,200	$1,200
4. Degree of operating leverage (row 2 ÷ row 3)	$\frac{\$3,200}{\$1,200} = 2.67$	$\frac{\$2,000}{\$1,200} = 1.67$	$\frac{\$1,200}{\$1,200} = 1.00$

- These results indicate that, when sales are 40 units, a 1% change in sales and contribution margin will result in 2.67% change in operating income for Option 1 and a 1% change in operating income for Option 3.
- Consider, for example, a sales increase of 50% from 40 to 60 units.

Operating Leverage

	Option 1	Option 2	Option 3
1. Contribution margin per unit (page 98)	$ 80	$ 50	$ 30
2. Contribution margin (row 1 × 60 units)	$4,800	$3,000	$1,800
3. Operating income (from Exhibit 3-5)	$2,800	$2,200	$1,800
4. Degree of operating leverage (row 2 ÷ row 3)	$\frac{\$4,800}{\$2,800} = 1.71$	$\frac{\$3,000}{\$2,200} = 1.36$	$\frac{\$1,800}{\$1,800} = 1.00$

- The degree of operating leverage decreases from 2.67 (at sales of 40 units) to 1.71 (at sales of 60 units) under Option 1 and from 1.67 to 1.36 under Option 2. In general, whenever there are fixed costs, the degree of operating leverage decreases as the level of sales increases beyond the breakeven point.

Effects of Sales Mix on Income

- The formulae presented to this point have assumed a single product is produced and sold
- A more realistic scenario involves multiple products sold, in different volumes, with different costs For simplicity's sake, only two products will be presented, but this could easily be extended to even more products.

	GMAT Success	*GRE Guarantee*	Total
Expected sales	60	40	100
Revenues, $200 and $100 per unit	$12,000	$4,000	$16,000
Variable costs, $120 and $70 per unit	7,200	2,800	10,000
Contribution margin, $80 and $30 per unit	$ 4,800	$1,200	6,000
Fixed costs			4,500
Operating income			$ 1,500

- What is the breakeven point for Emma's business now? The total number of units that must besold to break even in a multiproduct company depends on the sales mix.

Effects of Sales Mix on Income

- We assume that the budgeted sales mix (60 units of *GMAT Success* sold for every 40 units of *GRE Guarantee* sold, that is, a ratio of 3:2) will not change at different levels of total unit sales. That is, we think of Emma selling a bundle of 3 units of *GMAT Success* and 2 units of *GRE Guarantee*.

Each bundle yields a contribution margin of $300, calculated as follows:

	Number of Units of *GMAT Success* and *GRE Guarantee* in Each Bundle	Contribution Margin per Unit for *GMAT Success* and *GRE Guarantee*	Contribution Margin of the Bundle
GMAT Success	3	$80	$240
GRE Guarantee	2	30	60
Total			$300

Effects of Sales Mix on Income

To compute the breakeven point, we calculate the number of bundles Emma needs to sell.

$$\text{Breakeven point in bundles} = \frac{\text{Fixed costs}}{\text{Contribution margin per bundle}} = \frac{\$4{,}500}{\$300 \text{ per bundle}} = 15 \text{ bundles}$$

The breakeven point in units of *GMAT Success* and *GRE Guarantee* is as follows:

GMAT Success: 15 bundles × 3 units per bundle	45 units
GRE Guarantee: 15 bundles × 2 units per bundle	30 units
Total number of units to break even	75 units

The breakeven point in dollars for *GMAT Success* and *GRE Guarantee* is as follows:

GMAT Success: 45 units × \$200 per unit	\$ 9,000
GRE Guarantee: 30 units × \$100 per unit	3,000
Breakeven revenues	\$12,000

Effects of Sales Mix on Income

When there are multiple products, it is often convenient to use the contribution margin percentage. Under this approach, Emma also calculates the revenues from selling a bundle of 3 units of *GMAT Success* and 2 units of *GRE Guarantee*:

	Number of Units of *GMAT Success and GRE Guarantee* in Each Bundle	Selling Price for *GMAT Success* and *GRE Guarantee*	Revenue of the Bundle
GMAT Success	3	$200	$600
GRE Guarantee	2	100	200
Total			$800

Contribution Margin Versus Gross Margin

Gross margin = Revenues − Cost of goods sold

Contribution margin = Revenues − All variable costs

- The gross margin measures how much a company can charge for its products over and above the cost of acquiring or producing them.
- Contribution margin indicates how much of a company's revenues are available to cover fixed costs. It helps in assessing the risk of losses.
- Gross margin and contribution margin are related but give different insights. For example, a company operating in a competitive market with a low gross margin will have a low risk of loss if its fixed costs are small.

Contribution Margin Versus Gross Margin

Contribution Income Statement Emphasizing Contribution Margin (in thousands)			Financial Accounting Income Statement Emphasizing Gross Margin (in thousands)	
Revenues		$1,000	Revenues	$1,000
Variable manufacturing costs	$250		Cost of goods sold (variable manufacturing costs, $250 + fixed manufacturing costs, $160)	410
Variable operating (nonmanuf.) costs	270	520		
Contribution margin		480	Gross margin	590
Fixed manufacturing costs	160			
Fixed operating (nonmanuf.) costs	138	298	Operating (nonmanuf.) costs (variable, $270 + fixed, $138)	408
Operating income		$ 182	Operating income	$ 182